Project Design Phase-II Technology
Stack (Architecture & Stack)

Date	30 January 2026
Team ID	LTVIP2026TMIDS86494
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Technical Architecture:

Visualization tool for Electric consumption Analysis in 2019 and 2020s

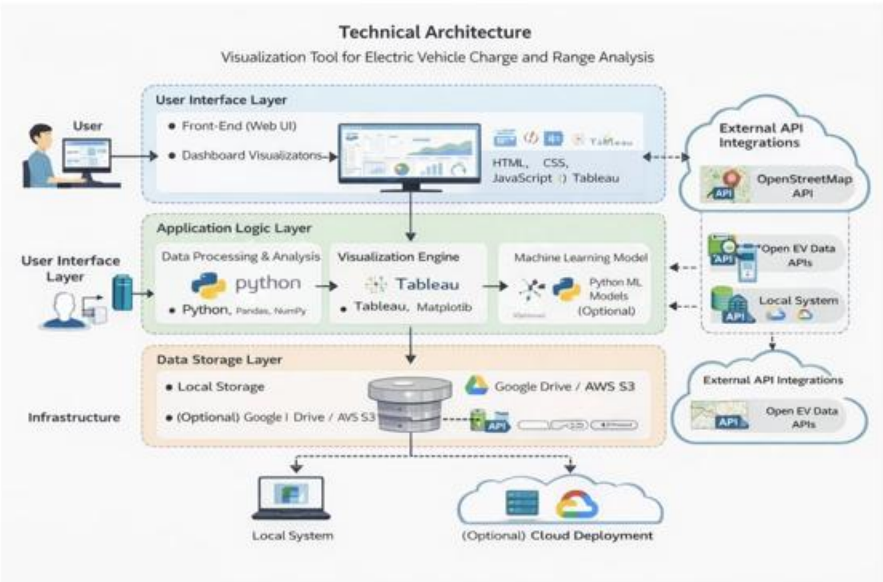

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface to view electric consumption visualizations with Tableau	HTML, CSS, JavaScript, Tableau
2.	Application Logic-1	Data processing and analysis	Python
3.	Application Logic-2	Data cleaning and transformation	Pandas, NumPy
4.	Application Logic-3	Visualization logic	Tableau, Matplotlib
5.	Database	Extracting from consumption file	CSV Files
6.	Cloud Database	Cloud storage for datasets	. Google Drive
7.	File Storage	Storage of reports and datasets	Local File System
8.	External API-1	Electric consumption data statewise	Open Electric consumption Data APIs
9.	External API-2	Geographic data for maps	Open in Tableau
10.	Machine Learning Model	Range estimation (optional)	Python ML Models
11.	Infrastructure (Server / Cloud)	Application deployment	Local System

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for development	Python, Tableau
2.	Security Implementations	Data access control and protection	File permissions, basic encryption
3.	Scalable Architecture	Supports adding more datasets and charts	Modular architecture
4.	Availability	System available during analysis time	Local
5.	Performance	Fast loading dashboards	Optimized queries, caching